

Module 5

Discrete-Time Fourier Transform (DTFT)

Section A — Basic Problems

Problem 1

Generate the discrete-time signal

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$

Plot the signal and identify:

- Amplitude
- Decay rate
- Nature of the signal

Problem 2

Compute and plot the DTFT magnitude spectrum of

$$x[n] = \delta[n]$$

Comment on the spectrum.

Problem 3

Generate

$$x[n] = \cos\left(\frac{\pi n}{4}\right)$$

Plot:

- Time-domain signal
- DTFT magnitude spectrum

Problem 4

Compare the DTFT spectra of:

- $x_1 = \cos(\pi n/8)$
- $x_2 = \cos(\pi n/2)$

Problem 5

Investigate the effect of increasing signal length on DTFT resolution.

Section B — DTFT Properties

Problem 6

Verify the Linearity Property.

Problem 7

Verify the Time Shifting Property.

Problem 8

Verify the Frequency Shifting Property.

Problem 9

Verify the Time Reversal Property.

Problem 10

Verify the Conjugation Property.

Section C — Programming Problems

Problem 11

Generate and analyze a rectangular sequence.

Problem 12

Compare the DTFT of:

- Rectangular sequence
- Triangular sequence
- Exponential sequence

Problem 13

Window a discrete-time signal and compare its spectrum.

Problem 14

Verify Circular Convolution using the DFT.

Problem 15

Compare DTFT and DFT of the same signal.

Challenge Problems

Challenge 1

Implement the DTFT manually without using FFT functions.

Challenge 2

Implement a simple spectral analyzer.

Challenge 3

Compare different window functions:

- Rectangular
- Hamming
- Hann
- Blackman

Challenge 4

Demonstrate spectral leakage.

Challenge 5

Investigate zero-padding and frequency resolution.

Challenge 6

Build a mini frequency detector that estimates the dominant frequency of an unknown discrete-time signal.